Fraud Detection System — Executive Summary

This project focused on building an advanced machine learning–based fraud detection system capable of identifying high-risk financial transactions in real time. The solution combined predictive modeling, explainable AI, interactive visualizations, and a Streamlit dashboard to simulate a production-level fraud analytics environment.

The final model was developed using the LightGBM algorithm and optimized through Optuna hyperparameter tuning. Among all evaluated models, LightGBM delivered the strongest performance with:

- Accuracy: 97.8%
- Precision: 91.4%
- Recall: 86.2%
- F1-Score: 88.7%
- PR-AUC Score: 0.91

The project emphasized PR-AUC rather than traditional accuracy because fraud datasets are highly imbalanced, with fraudulent transactions representing only a small percentage of total activity. The model demonstrated strong capability in detecting fraudulent behavior while minimizing false positives.

A multi-page Streamlit dashboard was developed to support fraud operations analysis. The dashboard included:

- Real-time fraud overview metrics
- Interactive transaction explorer
- Transaction-level risk scoring
- SHAP explainability visualizations
- Interactive Plotly charts and filters

Several business-focused visualizations were also created, including:

- Fraud rate by hour of day
- Transaction amount distribution
- Risk tier segmentation
- Precision-Recall analysis
- SHAP global feature importance
- Interactive fraud probability scatter plots

SHAP analysis identified the most influential fraud indicators:

- High transaction amount
- Suspicious late-night transaction timing
- Device and email risk signals

Critical-risk transactions commonly showed:

- Fraud probability above 80%
- Transaction amounts above $1,500
- Unusual device behavior
- Repeated transaction attempts during late-night hours

Based on model performance and fraud detection coverage, the system is estimated to prevent approximately:

- $180,000–$250,000 in annual fraudulent losses

Two major fraud prevention recommendations were proposed:

1. Real-time automated fraud flagging for high-risk transactions
2. Multi-factor authentication for suspicious device activity and high-value transactions

The project also highlighted several real-world machine learning challenges, including evolving fraud behavior, class imbalance, and feature limitations. Future improvements could include:

- IP geolocation intelligence
- Device fingerprinting
- Merchant risk profiling
- Behavioral analytics
- Real-time anomaly detection

Overall, this project demonstrates practical skills in:

- Machine Learning
- Fraud Analytics
- Explainable AI (SHAP)
- Streamlit Dashboard Development
- Interactive Data Visualization
- Model Evaluation
- Business Risk Analysis

The final solution was deployed on Streamlit Community Cloud and structured as a portfolio-ready end-to-end fraud analytics project.